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REVIEW

# Intravenous Amiodarone in Preexcited Atrial Fibrillation: A Systematic Review

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**ABSTRACT:** The current guidelines contraindicate intravenous amiodarone in patients with Wolff-Parkinson-White syndrome presenting with preexcited atrial fibrillation (AF) due to the risk of degeneration into ventricular fibrillation (VF). However, these recommendations are based predominantly on isolated case reports, which is concerning given the drug's widespread global use as a first-line antiarrhythmic in resource-limited settings. To evaluate the safety of intravenous amiodarone in this context, we conducted a systematic review of (1) studies enrolling patients with electrocardiographically confirmed preexcited AF who received intravenous amiodarone and (2) studies quantifying antegrade accessory-pathway effective refractory period during intravenous amiodarone administration. All observational and interventional studies assessing patients with preexcited AF were pooled under a conjugate β–binomial model with prespecified weak priors to estimate the risk of VF during or following infusion. Concomitantly, to assess real-world access to alternative class IIa/IIb European Society of Cardiology-recommended antiarrhythmic agents for preexcited AF, we performed a multinational survey of Latin American emergency departments. Twelve studies comprising 177 patients were included in the review (7 case reports, 2 cohorts, 3 before-and-after interventional studies). Four case reports described transient ventricular rate acceleration or VF following intravenous amiodarone. However, across the observational and interventional cohorts assessing patients with preexcited AF (n=146), no acceleration or VF was observed. The posterior mean estimates of VF risk ranged 0.12% to 0.68% over priors. Across 3 interventional before-and-after studies, there was a significant increase in the anterograde effective refractory period of the atrioventricular node and AP following amiodarone administration. The survey responses from 10 emergency centers indicated that none had access to class IIa/IIb indicated agents, while all had intravenous amiodarone available. Taken together, population-level evidence suggests that the risk of VF in preexcited AF following intravenous amiodarone administration is rare. In settings where guideline-recommended drugs are inaccessible, intravenous amiodarone may represent a clinically reasonable alternative for rhythm or rate control.

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**Key Words:** acceleration ■ amiodarone ■ atrial fibrillation ■ atrioventricular node ■ ventricular fibrillation

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Wolff-Parkinson-White (WPW) syndrome is a congenital disorder characterized by the presence of extra-atrioventricular connection(s) that may facilitate preexcitation of the ventricles. This syndrome often predisposes patients to episodes of atrial fibrillation (AF),[1–4] which is observed in up to 50% of patients with WPW across their lifespan.[4–6] The presence of preexcitation during AF creates a substrate for rapid ventricular conduction via the accessory pathway (AP) due to its often-nondecremental conduction, particularly in the setting of short anterograde effective refractory periods (AERPs).[5–8s]

The guideline-recommended approach for the acute management of preexcited AF is direct current cardioversion, especially in cases of hemodynamic instability.[7,8] However, in stable patients or situations in which we attempt to reduce the failure rate from immediate preexcited AF recurrence following shock, pharmacological cardioversion is either attempted first or given as a pretreatment. In these circumstances, drugs that slow atrioventricular conduction, such as beta and calcium-blockers, have the potential to be harmful.[7,8] Atrioventricular nodal blockade causes atrial impulses to be preferentially conducted via the AP, leading to rapid

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## Nonstandard Abbreviations and Acronyms

| | |
|---|---|
| **AERP** | anterograde effective refractory period |
| **AF** | atrial fibrillation |
| **AP** | accessory pathway |
| **AVN** | atrioventricular node |
| **ICa** | calcium channel |
| **INa** | sodium channel |
| **SPRRI** | shortest preexcited R-R interval |
| **VF** | ventricular fibrillation |
| **VR** | ventricular rate |
| **WPW** | Wolff-Parkinson-White |

ventricular responses that can degenerate into ventricular fibrillation (VF).[7–11]

In the past, amiodarone was the preferred drug of choice by the American Heart Association/American College of Cardiology for the management of preexcited AF, particularly when the rhythm could not be converted or ablated.[12] However, concerns arose due to case reports published from 1982 to 2012, describing acceleration of ventricular rate (VR) or degeneration into VF shortly after administration of intravenous amiodarone.[13–16] These concerns prompted revisions to international guidelines. The 2019 American Heart Association/American College of Cardiology consensus classified intravenous amiodarone as potentially harmful in patients with preexcited AF (class III recommendation, level B-NR),[8] and the 2019 European Society of Cardiology guidelines similarly contraindicate its use in this context (recommendation: class III, level of evidence: B).[7]

Nevertheless, in many parts of the world, amiodarone may be the only intravenous antiarrhythmic available for the acute treatment of preexcitation syndromes. Although the use of intravenous amiodarone in this context is an important decision, the supporting evidence for its contraindication in current guidelines is based on the lowest form of evidence,[7,8] and uncertainty remains due to novel studies demonstrating a lack of danger for arrhythmic sequelae.[17–19] In light of this, we conducted a comprehensive review of published reports assessing the safety and response rate of intravenous amiodarone in preexcited AF. As a secondary objective, we surveyed emergency rooms across Latin America to evaluate the availability of clinically recommended intravenous antiarrhythmic therapies for preexcited AF.

# METHODS

## Systematic Literature Review

An electronic literature review of the published data was conducted in MEDLINE and EMBASE databases. Two study populations were targeted. The first population comprised patients with WPW who presented to the emergency department or an inpatient ward with preexcited AF confirmed on ECG and were treated with intravenous amiodarone for acute management. The second population comprised patients with overt or concealed AP undergoing programmed electrophysiology testing who received protocolized intravenous amiodarone with AERP measured for the atrioventricular node (AVN) and AP both before and after infusion.

The selection of articles was made according to the following criteria: (1) publication date from database inception until October third, 2025; (2) human participants; (3) enrollment of at least one of the prespecified study populations; (4) availability of full-text article, and (5) primary research using a case-report, observational, or interventional design. Studies were excluded if amiodarone was administered exclusively via the oral route; however, studies were retained if oral administration was concurrent with intravenous therapy, which is common clinical practice.

Medical Subject Headings (MeSH)/Embase Subject Headings (Emtree) terminology and relevant keywords were applied to develop the literature search. Keywords used to search for articles pertaining to preexcited AF included: Preexcited Atrial Fibrillation (Keyword Term), WPW (Keyword Term), Wolff-Parkinson-White Syndrome (MeSH term; used when searching MEDLINE), Wolff Parkinson White Syndrome (Emtree; used when searching EMBASE). Literature about amiodarone was identified using the term: Amiodarone (MeSH/Emtree term).

A total of 863 articles were imported into the Covidence program for screening, and 83 duplicates were automatically removed. The abstracts of the remaining articles underwent independent review by 2 investigators (Z.A. and D.N.), and 764 were excluded due to not meeting eligibility criteria. During full-text screening, 4 additional studies were excluded due to their having administered only oral amiodarone, resulting in the selection of 12 studies. The results of the literature search are illustrated in Figure 1.

## Risk of Bias Assessment

The methodological quality of the included articles was assessed using the risk of bias in nonrandomized studies of interventions tool (ROBINS-I) for all observational and interventional studies, and the modified Newcastle-Ottawa Scale for all case reports. The ROBINS-I tool's items are classified into 7 domains: confounding, selection bias, intervention classification, deviations from intended interventions, missing data, outcome measurement, and selection of reported results.[20] The modified Newcastle-Ottawa Scale items are classified into 4 domains: selection, ascertainment, causality, and reporting.[21] Two investigators (Z.A. and A.A.) appraised each study independently, and discrepancies were resolved by consensus. Each domain in the included studies was categorized as having a low, moderate, serious, critical, or unclear (no information) bias.

## Statistical Analysis

The primary end point was the occurrence of VF or VR acceleration at any time during or shortly after administration of intravenous amiodarone in patients with preexcitation. The

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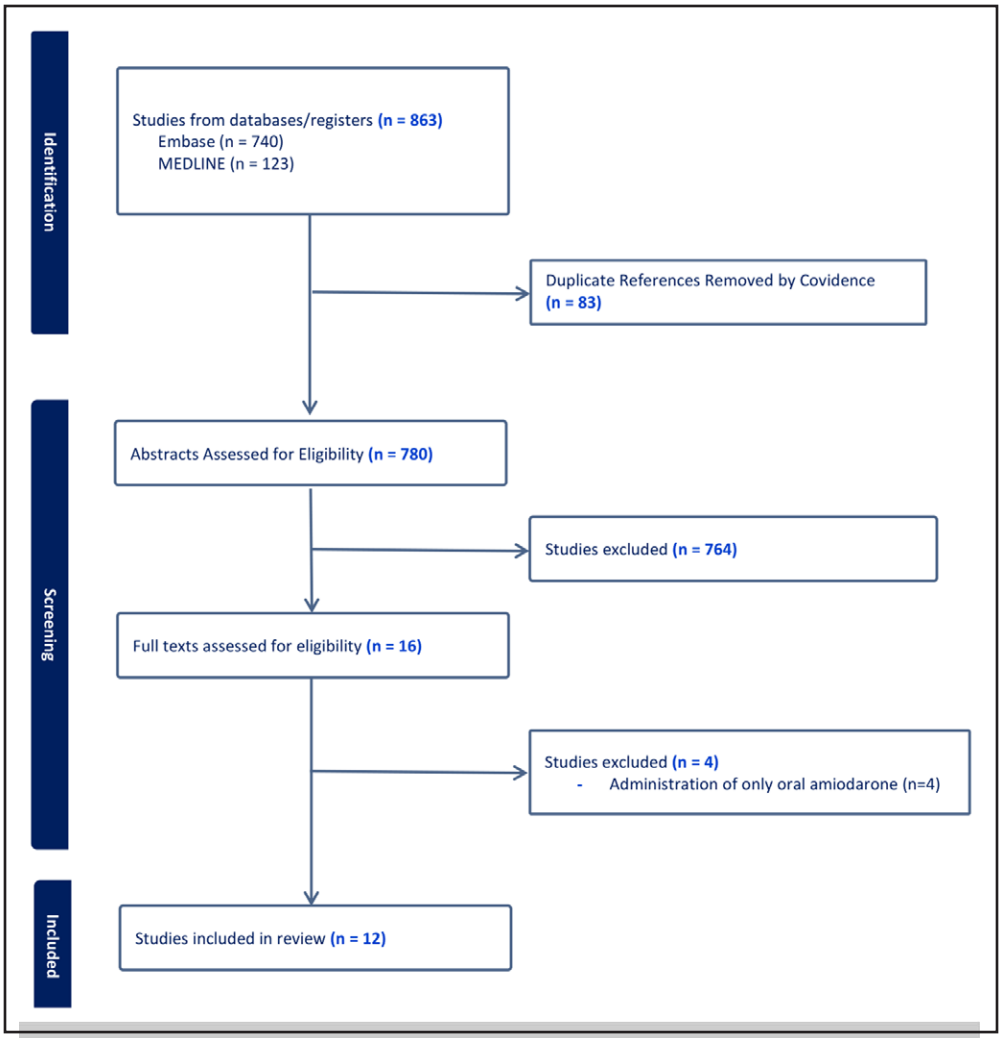


**Figure 1. Flow diagram of the method used for study selection.**

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patient-level risk $p$ of a VF or rate acceleration event was estimated across the observational and interventional studies. Assuming exchangeability for this acute end point, each patient contributed a single Bernoulli outcome (VF: yes/no) within that study's predefined observation window; with $k$ events among $n$ exposed patients, the likelihood is Binomial $k \mid n, p \sim \text{Binomial}(n, p)$.

We placed a $\text{Beta}(\alpha, \beta)$ prior on $p$ because the beta distribution is the conjugate prior for a Binomial proportion. This yields the closed-form posterior:

$$p \mid data \sim Beta(\alpha + k, \beta + n - k)$$

where, given the paucity of events, we adopted a fully Bayesian analysis and prespecified a small panel of weak-to-skeptical beta priors ($\alpha$ and $\beta$) to quantify the posterior probability (conditional on the accrued sample) that the true risk lies below clinically relevant thresholds.

The beta–binomial model was selected here because, in sparse-event settings, conventional random-effects meta-analysis of single-arm proportions may be weakly identified on the between-study heterogeneity ($\tau^2$) and yield pooled estimates that are sensitive to modeling choices (eg, transformations or continuity corrections) used to stabilize inference.

## Survey of Antiarrhythmic Availability in Latin America

A survey was developed in English to assess the availability of class IIa, IIb, and III drugs recommended by the European Society of Cardiology for the acute treatment of preexcited AF. This included intravenous ibutilide, intravenous procainamide, intravenous propafenone, intravenous flecainide, and intravenous amiodarone.[4] The survey was translated into Spanish by JELR and distributed by email to physician contacts at 27 hospitals in Argentina, Peru, Costa Rica, and Bolivia between May 2024 and June 2024. Responses were collected electronically and were used to characterize emergency room access to these agents at participating centers. This survey was designed as a regional assessment of drug availability using a convenience sampling frame and was not intended to estimate worldwide availability.

# RESULTS

## Study Characteristics

The main characteristics and findings from the 10 studies, involving 153 patients, presenting with preexcited AF

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who received intravenous amiodarone are summarized in Table 1. These studies span the time frame of 1982 to 2023. These articles include 7 case reports,[13–16,22–24] 2 single-arm observational studies,[17,18] and 1 single-arm before-and-after interventional study.[19] An additional 2 single-arm before-and-after interventional studies, spanning 1984 to 1987, presented patients with AP who received protocolized intravenous amiodarone and underwent electrophysiology testing.[25,26]

## Quality Assessment: ROBINS-I

The risk of bias for all interventional and observational studies according to the *ROBINS-I* tool is shown in Figure 2.

For the clinical safety outcome, the risk of confounding was serious for all studies. All cohorts assessed treatment-only and lacked concurrent comparators (single-arm); amiodarone was administered alongside

**Table 1. Arrhythmic Outcomes After the Administration of Intravenous Amiodarone**

| Authors | Study type | No. of patients | Age, y | Dosage | Rhythm at presentation | Most common location of AP | Outcome in conversion |
|---|---|---|---|---|---|---|---|
| Alizadeh et al[19] | Single-Arm Interventional Study | 103 | 34.67 | 300 mg in 30 min | AF, — | Right posteroseptal in 37 patients, left lateral in 36 patients | Amiodarone infusion restored 19 patients to SR (18%). Any patient requiring cardioversion following infusion was excluded. No ventricular arrhythmia was detected. |
| Acharya et al[17] | Single-Arm Retrospective and Prospective Cohort Study | 13 | 38.5±12.3 | … | AF, — | Right posteroseptal, coronary sinus epicardial | Amiodarone infusion restored 11 patients to SR (85%). Cardioversion shock following amiodarone restored 2 patients to SR. No ventricular arrhythmia was detected. |
| Ren et al[18] | Single-Arm Retrospective Cohort Study | 30 | 47.8±17.0 | Bolus of 150 to 300 mg in 10 to 20 min, followed by infusion of 1 mg/min | AF, mean ventricular rate 171 (150–189) bpm | … | Amiodarone infusion restored 18 patients to SR (60%). Cardioversion shock following amiodarone restored 11 patients to SR (37%). 1 patient failed to convert to SR. No ventricular arrhythmia was found. |
| Schützenberger et al[13] (E; A)* | Case Report | 1 | 60 | Bolus of 5 mg/kg in 10 min | AF, ventricular rate 140 bpm | … | The ventricular rate accelerated to 210 bpm within 10 min after amiodarone infusion. The rise in HR persisted for 10 min before SR was restored. |
| Sheinman and Evans[14] (E; A)*,† | Case Report | 1 | 69 | 800 mg by mouth, followed by infusion of 1200 mg in 24 h | AF, ventricular rate 130 to 170 bpm | … | The ventricular rate accelerated to 230 bpm several hours after amiodarone infusion began. 150 mg IV disopyramide restored SR. |
| Boriani et al[15] (E; A)* | Case Report | 1 | 32 | Bolus of 5 mg/kg in 20 min | AF, ventricular rate 250 bpm | Left posteroseptal | The ventricular rate degenerated to VF within 10 min after the amiodarone bolus. Cardioversion shock restored SR. |
| Leiria et al[16] | Case Report | 1 | 31 | Bolus of 150 mg in 10 min, followed by 1 mg/min | AF, ventricular rate 157 bpm | Left lateral | The ventricular rate degenerated to VF within minutes after the amiodarone bolus. Cardioversion shock restored SR. |
| Panduranga et al[22] | Case Report | 1 | 10 | Bolus of 40 mg (1 mg/kg) in 10 min, followed by 100 mg with cardioversion. | AF, ventricular rate 200 to 350 bpm | Left posteroseptal | Amiodarone infusion had a transient response. Electrical cardioversion shock restored SR. |
| Payami et al[23] | Case Report | 1 | 34 | Bolus of 150 mg in 10 min repeated twice | AF, ventricular rate 250 bpm | Right posteroseptal | Amiodarone infusion restored SR. |
| Takahira et al[24] | Case Report | 1 | 66 | Bolus of 125 mg in 10 min, followed by 300 mg in 6 h, then 450 mg in 18 h | AF, ventricular rate 250 bpm | Left anterolateral | Amiodarone infusion decreased HR to 80 bpm with no ventricular preexcitation. |

Studies with E are included in the current ESC guidelines, and A in the current ACA/AHA guidelines. ACA indicates American College of Cardiology; AF, atrial fibrillation; AHA, American Heart Association; AP, accessory pathway; BPM, beats per minute; ESC, European Society of Cardiology; HR, heart rate; SR, sinus rhythm; and VF, ventricular fibrillation.

*Included in a case series cited by the ACA/AHA.

†Included in a case series cited by the ESC.

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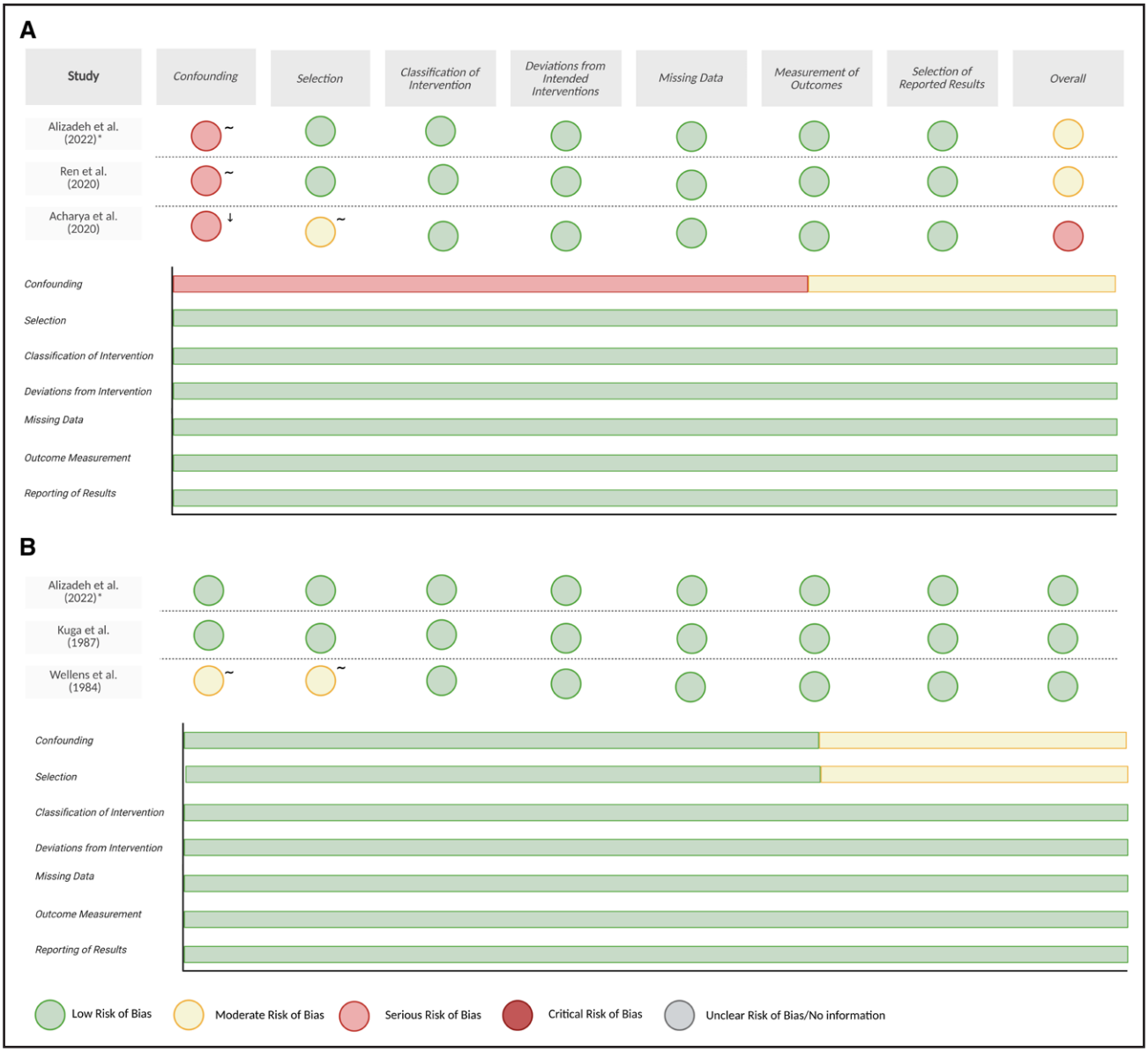


**Figure 2. ROBINS-I risk-of-bias assessments for nonrandomized studies.**
**A**, Observational and interventional studies assessing clinical safety outcomes (ventricular fibrillation [VF] or ventricular-rate acceleration) in patients with preexcited atrial fibrillation (AF) following intravenous (IV) amiodarone infusion. **B**, Interventional studies assessing AERP in patients with Wolff-Parkinson-White (WPW) before and after IV amiodarone infusion. ↓ Bias directed towards null, ↑ bias directed away from null; ≈ unknown direction of bias. *Alizadeh et al[19] is assessed separately based on the population of interest.

cointerventions (eg, sedation, potassium/magnesium supplements); and no study adjusted for measured confounders. Selection bias was low in Ren et al[18] and Acharya et al[17] (time-delimited census of all treated patients), but moderate in Alizadeh et al[19] owing to exclusion of patients requiring cardioversion within 30 minutes of infusion (an effect that may be influenced by the intervention).

For the electrophysiology outcome, the risk of confounding was low–to–moderate. The paired before–and–after electrophysiology protocols and brief assessment windows minimized baseline confounders and time-varying confounders, respectively. Antiarrhythmic washout was reported in Alizadeh et al[19] and Kuga et al,[26] but not in Wellens et al.[25] It is important to note that both Alizadeh et al and Wellens et al used convenience electrophysiology lab samples.

## Quality Assessment: Modified Newcastle-Ottawa

The risk of bias for all case reports according to the Modified Newcastle-Ottawa tool is shown in Figure 3. The causality bias was serious to critical in all cases: alternative explanations were incompletely addressed (lack of acknowledgment of coadministered antiarrhythmic and

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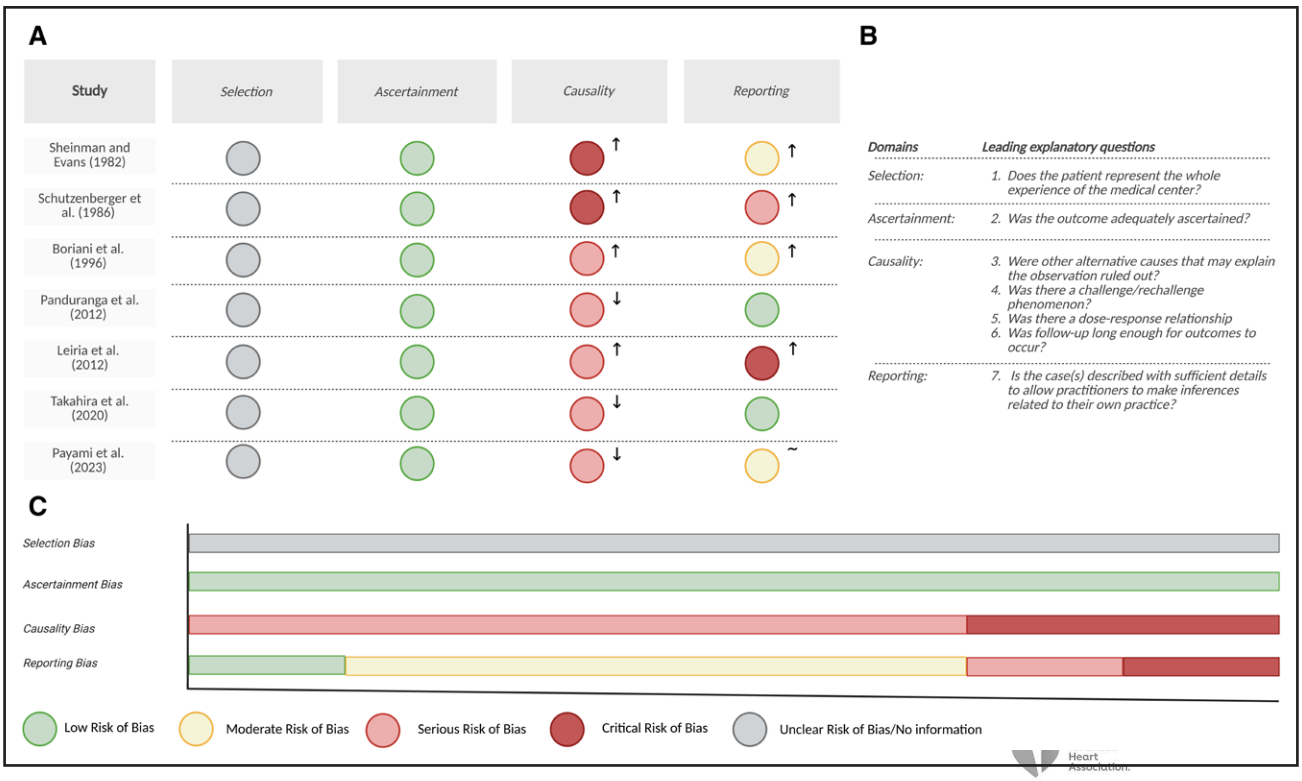


**Figure 3. Modified Newcastle–Ottawa assessment of case reports.**
**A**, Domain ratings for 7 case reports. **B**, Domain anchors used for adjudication (lead questions shown). **C**, Summary distribution of risk across domains. ↓, bias directed towards null; ↑, bias directed away from null; ≈, unknown direction of bias.

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family/patient history on outcome), and none demonstrated dose–response, or (re)challenge.

## Case Reports

Two reports, by Schützenberger et al[13] and Scheinman and Evans, documented significant VR acceleration following amiodarone infusion. Schützenberger et al[13] described a patient with a longstanding history of recurrent preexcited AF and rapid ventricular responses. During one spontaneous episode (initial VR, ≈140 bpm; shortest preexcited R-R interval [SPRRI], 300 ms), intravenous ajmaline, propafenone, and direct current shock failed to restore sinus rhythm. Thus, a bolus of intravenous amiodarone was administered, after which the VR increased to 210 bpm (SPRRI, 240 ms) and the patient developed hypotension (SBP, 80 mmHg) within 10 minutes.[13] Similarly, Scheinman and Evans reported a patient with a history of myocardial infarction, presenting with left ventricular failure and irregular tachycardia (130–170 bpm) refractory to verapamil, disopyramide, and lidocaine.[14] They administered amiodarone by mouth followed by a continuous infusion, and several hours later, the patient's rhythm deteriorated into broad-complex AF with a peak VR of 230 bpm.[14]

Two additional cases, by Boriani et al[15] and Leiria et al, described degeneration of AF into VF following amiodarone infusion. Boriani reported a patient admitted due to a syncopal episode with AF and rapid ventricular preexcitation, at a mean rate of 250 bpm (SPRRI of 190 ms). The patient received a bolus of amiodarone over 20 minutes, and a few minutes after its termination, collapsed with VF.[15] Leiria et al[16] reported a similar clinical trajectory in a hemodynamically stable patient with AF and irregular wide-complex tachycardia (heart rate=157 bpm). Following a 150 mg bolus of amiodarone over 10 minutes, the patient progressed to VF.[16]

In contrast, 3 case reports described administration of intravenous amiodarone without any evidence of proarrhythmic events. In the study by Payami et al,[23] a patient presented with AF and a pulse rate of 250 bpm (SPRRI of 220 ms). The patient was administered an amiodarone loading dose over 10 minutes, repeated twice due to a lack of response to the first bolus.[23] SR was observed within 4 minutes of administration.[23] In Takahira et al,[24] a patient was described with decompensated heart failure (systolic dysfunction) causing pulmonary congestion, and preexcited AF (presenting rate=170 bpm, SPRRI=380 ms), in whom electrocardioversion was deferred due to a left atrial thrombus. The patient was instead treated with a bolus of intravenous amiodarone over 10 minutes, followed by an infusion for 24 hours.[24] The rhythm converted to SR without complications.[24]

## Clinical Studies: Safety Outcomes

Ren et al[18] conducted a single-arm, single-center retrospective analysis of 30 patients who presented to the

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emergency room with electrocardiographically confirmed preexcited AF and received intravenous amiodarone. The patients were relatively stable at admission (heart rate=150–189 bpm); however, a substantial portion had comorbidities (47%).[18] The majority of patients (60%) converted to sinus rhythm pharmacologically, while an additional 11 (37%) required direct current cardioversion (10 at physician determination and 1 at development of severe hypotension). No cases of VR acceleration or VF were observed during or after infusion. No other antiarrhythmic agents were used in patients, apart from propafenone in 2 cases.[18]

Similarly, Acharya et al reported on a combined single-arm retrospective and prospective cohort of 41 patients presenting with WPW syndrome and preexcited AF.[17] Among these, 13 hemodynamically stable patients received intravenous amiodarone as first-line therapy. Sinus rhythm was restored in 11 patients, while the remaining 2 developed hypotension and were electrically cardioverted. No cases of ventricular arrhythmias were observed.[17]

Lastly, Alizadeh et al[19] performed a single-arm before-and-after interventional study to evaluate the electrophysiological effects of intravenous amiodarone in 103 patients with either spontaneous or accidentally induced preexcited AF. All antiarrhythmic medications were withheld for at least 5 half-lives before admission. Although only 19 patients (18.4%) achieved pharmacological cardioversion, the intervention was not associated with any instance of VR acceleration or VF during the monitoring period (2 hours).[19]

## Beta-Binomial Estimated Risk

Across the pooled observational and interventional evidence (n=146 exposed to intravenous amiodarone), no instances of VR acceleration or degeneration into VF were observed (k=0). The intravenous dosing regimens in these studies were broadly consistent with contemporary clinical practice (although not explicitly reported in Acharya et al), whereas the greatest variability arose from patient selection (intervention versus cohort design) and baseline clinical characteristics (notably hemodynamic stability and comorbidity burden).

To estimate the event risk compatible with observing no events in this finite sample, we used a conjugate beta-binomial model (under weak priors) to quantify the range of event probabilities and to derive posterior upper credible bounds on risk given the available evidence. The posterior means and upper bounds were of similar magnitude across weak priors (Table 2). The posterior probability that the true risk is ≤1% ranged from 0.77 (uniform) to 0.98 (skeptical).

## Clinical Studies: Electrophysiological Outcomes

Three interventional, single-arm studies evaluated the effect of intravenous amiodarone on the AERP of the AVN and AP.[19,25,26] All studies determined AERP using programmed atrial extrastimuli (S1–S2) delivered on a fixed basic drive. All studies reported statistically significant prolongation of both AVN and AP-AERP following amiodarone administration.

Alizadeh et al[19] administered a 300 mg intravenous amiodarone infusion over 30 minutes.[19] During basic atrial pacing at an undefined cycle length, the mean AERP prolongation was 41.95±26.36 ms (n=87, $P<0.001$) for the AVN and 22.30±30.90 ms (n=103, $P<0.001$) for the AP.

Wellens et al[25] administered a 5 mg/kg intravenous amiodarone infusion over 1 minute. During basic atrial pacing at a cycle length of 600 ms, the AVN AERP increased from 287±31 ms to 326±35 ms (mean prolongation: 39 ms, n=8; $P<0.05$), and the AP AERP increased from 265±14 ms to 305±21 ms (mean prolongation: 40 ms, n=9; $P<0.05$).[25]

Kuga et al[26] infused 5 mg/kg intravenous amiodarone over 5 minutes. During basic atrial pacing at a cycle length of 667 ms, the AVN AERP increased from 268±47 ms to 315±65 ms (mean prolongation: 47 ms, n=13; $P<0.01$), and the AP AERP increased from 288±34 ms to 316±58 ms (mean prolongation: 28 ms, n=15; $P<0.05$).[26]

## Survey Results

Table 3 summarizes the survey results. Of 27 solicited centers, 10 responded (response rate: 37%); 8

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**Table 2. Bayesian Beta-Binomial Estimation of Ventricular Fibrillation Risk After Intravenous Amiodarone**

| Prior | Before posterior distribution | Prior mean | Posterior mean | 95% CrI (2-sided) | Probability of ≤0.5% risk | Probability of ≤1% risk |
|---|---|---|---|---|---|---|
| Uniform | Beta (1–1) to beta (1–147) | 50 % | 0.68 % | 0.02%–2.48% | 0.52 | 0.77 |
| Jeffreys | Beta (0.5–0.5) to beta (0.5–146.5) | 50 % | 0.34 % | ≈0.00%–1.70% | 0.77 | 0.91 |
| Skeptical | Beta (0.2–19.8) to beta (0.2–165.8) | 1.0 % | 0.12 % | ≈0.00%–0.91% | 0.93 | 0.98 |

Before posterior distribution shows the beta prior updated by the observed evidence (k events among n exposed) to yield the posterior beta distribution. Posterior mean is the expected event risk under the posterior. 95% CrI is the interval containing 95% of the posterior probability for the true risk. The probability of ≤0.5% risk and the probability of ≤1% risk are the posterior probabilities that the true risk is below these prespecified thresholds. CrI indicates credible interval.

**Table 3. Availability of Amiodarone in the ER Among Responding Latin American Centers**

| Location | Center | Public/private | Available drugs in ER |
|---|---|---|---|
| Tucuman, Argentina | Centro Integral Arritmias Tucumán | Private | Amiodarone was the only drug available in the ER. |
| Juliaca, Peru | Hospital Carlos Monge Medrano | Public | Amiodarone was the only drug available in the ER. |
| Juliaca, Peru | Hospital Base III Juliaca | Private | Amiodarone was the only drug available in the ER. |
| San José, Costa Rica | Hospital México | Public | Amiodarone was the only drug available in the ER. |
| Buenos Aires, Argentina | Hospital Argerich | Public | Amiodarone was the only drug available in the ER. |
| La Paz, Bolivia | Instituto Nacional de Torax | Public | Amiodarone was the only drug available in the ER. |
| Buenos Aires, Argentina | Hospital Ramos Mejía | Public | Amiodarone was the only drug available in the ER. |
| Piura, Peru | Hospital de la Amistad Perú–Corea, Santa Rosa | Public | Amiodarone was the only drug available in the ER. |
| Lima, Peru | Hospital Nacional Daniel Alcides Carrion | Public | Amiodarone was the only drug available in the ER. |
| Chancay, Peru | Hospital Hidalgo Atoche | Public | Amiodarone was the only drug available in the ER. |

Solicited nonresponding centers are not recorded. ER indicates emergency department.

were public, and 2 were private centers across Argentina, Peru, Costa Rica, and Bolivia. Responding centers reported no emergency room availability for the first- or second-line agents endorsed by European Society of Cardiology guidelines (class IIa indication: procainamide, ibutilide; class IIb indication: flecainide, propafenone, sotalol) for acute management of preexcited AF. However, all responding centers indicated that intravenous amiodarone was available in the emergency room.

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## DISCUSSION

This review sought to evaluate the safety of amiodarone in cases of preexcited AF. The current European Society of Cardiology and American College of Cardiology/American Heart Association guidelines for the management of preexcited AF contraindicate the use of amiodarone in such situations due to the risk of precipitating VF.[7,8] However, our results suggest that intravenous amiodarone may be less dangerous than current guidelines imply for rhythm control in this patient population, despite the few case studies reporting adverse outcomes.

### Clinical Overview

Although the evidence points toward amiodarone being relatively safe for rhythm control in preexcited AF, the few reported instances of increased VR should not be overlooked. These cases may be attributable to 2 instances: (1) confounding and (2) characteristic differences in the AP.

#### *Confounding From Polypharmacy in Case Reports*

Firstly, confounding variables within the case studies are present. In Schützenberger et al,[13] the class Ia and class Ic intravenous drugs ajmaline (class Ia) and propafenone (class Ic) were administered beforehand, and the time between these treatments and amiodarone administration was not reported. Similarly, in Scheinman and Evans, the patient had failed to respond to intravenous verapamil (class IV), disopyramide (class Ia), and lidocaine (class Ic) before amiodarone administration.[14]

Administration of verapamil in the latter case may cause severe depression of the AVN when used in combination with amiodarone, favoring conduction over the AP.[7,8,27,28] The concurrent administration of class I antiarrhythmics and amiodarone in both cases may cause severe blockade of the sodium channel, precipitating monomorphic ventricular tachycardia due to the reduced conduction velocity that promotes reentry.[29] This is particularly in the presence of fibrosis,[29] as in the Scheinman and Evans case, where the patient had a prior myocardial infarction.[14] Moreover, a nonresponse to sequential class I antiarrhythmics before amiodarone in both cases implies an AP and atrial substrate with relative insensitivity to sodium-channel ($I_{Na}$) blockade, potentially attenuating the class I-like component of amiodarone's acute action on the AP.

The polypharmacy may also contribute to hemodynamic instability. The case of Schützenberger et al, but not Scheinman and Evans, reported hypotension; whether blood pressure was monitored in Scheinman and Evans is unclear. This effect is likely due to the negative inotropic actions of disopyramide and lidocaine, conjointly with the peripheral vasodilation from verapamil

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and polysorbate 80, an emulsifier in commercial intravenous amiodarone formulations known to transiently reduce mean arterial pressure.[30] Such acute hypotension can trigger compensatory baroreceptor reflexes, leading to beta-adrenergic stimulation that shortens the AP-ERP and ventricular action potential duration.[14] A similar adrenergic hyperstimulation may have been initiated by syncope in the case of Boriani et al.

### *Characteristics of the Accessory Pathway*

The anatomic location of the AP does not appear to account for the heterogeneity in response to intravenous amiodarone. Across reports, we observed no consistent trend between AP location and the prolongation of AERP or VF response. Many researchers believe that VF induction following amiodarone infusion requires an AP with critically short antegrade refractoriness,[19,22,23] for which the shortest preexcited RR interval in the emergency setting is often used as a surrogate indicator (commonly <250 ms).[1,31] However, the available evidence does not show that low shortest preexcited RR interval or low AP-AERP predicts amiodarone-associated VF or ventricular-rate acceleration: cases reporting VR acceleration span shortest preexcited RR interval values from 190 to 300 ms,[13,14] whereas low shortest preexcited RR interval values (<220 ms) and rapid baseline VRs are observed in cases where amiodarone successfully achieved rhythm control.[18,21]

These observations are concordant with electrophysiology data. Wellens et al[32] demonstrated that among the intravenous antiarrhythmics—procainamide, quinidine, ajmaline, and amiodarone—amiodarone was the only drug to produce a significant increase in the AP-AERP in patients with a baseline AERP of 270 ms or less (mean increase of 77.7 ms). Among this cohort, amiodarone produced lengthening of the AP-AERP in all cases included.[32] Although this increase was modest compared with patients with an initial AERP >270 ms (mean >187.5), such lengthening would result in a decrease in maximal VR during AF from 300 to near 220 bpm.

## Molecular Overview

Amiodarone possesses the properties of all 4 Singh-Vaughan Williams classes of antiarrhythmic agents.[33–35] Studies debating the safety of intravenous amiodarone in cases of preexcited AF suggest that its peculiar combination of electrophysiological effects may alter atrioventricular nodal properties with minimal impact on fast channel tissues.[15,16,36] Many have attributed this effect to the calcium channel blocking activity of amiodarone, along with its rate-dependent phenomenon.[7,8,15] This may be synergistic with amiodarone's anti-β-adrenergic effects, slowing conduction over the AVN and decreasing concealed retrograde conduction into the AP by normally conducted beats, causing preferential conduction down the AP and thereby increasing the VR.[7,8,15] However, these statements contradict both clinical and experimental observations.

### *Clinical Observations*

The consistent prolongation of refractoriness among the interventional electrophysiology studies measuring the AERP following amiodarone infusion indicates persistent increases of the SPRRI in patients, slowing the VR.[19] No study from our search demonstrated shortening or nonsignificant lengthening of the AP on amiodarone administration. Such results suggest that amiodarone is unable to accelerate the VR by increasing conduction through the AP. However, the more substantial prolongation of AVN than AP refractoriness may lead to wider QRS morphologies, causing a worsening preexcitation index.[19]

### *Experimental Observations*

These clinical findings for AP refractoriness are consistent with experimental evidence. Acute amiodarone exerts class I and IV properties, inhibiting the fast sodium current ($I_{Na}$) and slow calcium current ($I_{CaL}$).[35,37–39] At therapeutic concentrations of the drug ex vivo, the percent inhibition of the maximum rate of rise (Vmax) during use-dependent blocks is similar between fast and slow action potentials.[37–48] The pharmacological potency of amiodarone correlates well with its binding affinity to its receptors.[35] Consequently, IC50 values derived from radioligand assays indicate that at the concentrations of amiodarone seen in the myocardium following infusion, it would be expected to inhibit both $I_{Na}$ and $I_{CaL}$ channels in vivo as well.[36,49,50]

In addition to inhibiting Vmax, amiodarone's block of inactivated $I_{Na}$ and $I_{CaL}$ channels prolongs the AERP of both myocardial, inducing postrepolarization refractoriness,[37,51] and nodal tissue, respectively.[44] Both phenomena stem from delayed recovery of excitability, with time constants of recovery ($\tau_{rec}$) for drug-bound $I_{CaL}$ and $I_{Na}$ channels being close at 1080 ms and 1480 to 1630 ms.[38,40,47] This suggests that the re-activation window of both the AP and AVN is shifted to a positive direction, similarly during preexcited AF, with the greater prolongation of the AVN seen in the clinical studies likely being attributed to the synergistic anti-β-adrenergic effects on the AVN.

### *Risk of 1:1 Conduction Due to Amiodarone*

The occurrence of VF with intravenous amiodarone administration may be independent of an action on the AVN. The suggestion that amiodarone increases the VR by selectively slowing atrioventricular nodal conduction is unlikely, as previously discussed. Even if this mechanism were valid, it would not significantly affect patients with rapid ventricular responses during AF, where most impulses are already conducted via the AP, as documented in the case studies.

Amiodarone may instead coarsen the fibrillatory waves and prolong the AF cycle length near the AP-AERP. The

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AF cycle length serves as a surrogate marker for local refractoriness.[52] Amiodarone has been previously shown to increase the AF cycle length across multiple measured atrial sites, as well as significantly decrease AF cycle length dispersion (24±10 msec after cardioversion to 15±11 msec after amiodarone therapy with repeat cardioversion).[53] Consequently, the atrial arrhythmia may slow sufficiently for 1:1 conduction down the AP to become possible (with the anti-β-adrenergic effects likely preventing such an effect down the AVN). In Scheinman and Evans, the VR acceleration was several hours after amiodarone infusion (rather than immediately),[14] supporting this mechanism, and precluding the above mechanisms due to polypharmacy. This electrophysiological phenomenon has previously been described for class Ic antiarrhythmic agents in preexcitation cases, including both flecainide and propafenone.[54–56]

Intravenous propafenone has been documented to enhance atrioventricular conduction during atrial flutter in WPW syndrome, as observed during an electrophysiological study.[55] The VR acceleration was paralleled by an increase in the atrial flutter cycle length. However, cohort studies have demonstrated no significant risks of intravenous propafenone in patients with preexcited AF.[57,58] Similarly, flecainide has been reported to cause sudden death in a patient with WPW and paroxysmal AF.[56] These observations suggest that agents with potent block of the inactivated sodium channel, including class Ic agents and possibly amiodarone, may cause rare pro-arrhythmic events in situations of preexcited AF; a well-documented behavior that is long recognized across the AVN in normal atrial flutter/fibrillation.

This effect across the AP in preexcited AF/flutter is primarily a result of the prolonged $\tau_{rec}$ and rate-dependence of the class I agents,[59] which allow them to smooth out differences in ERP at rapid heart rates for atrial tissues with inhomogeneity. However, amiodarone's $\tau_{rec}$ more closely resembles those of class Ia agents, such as procainamide.[38,40,47] As a result, the likelihood of amiodarone inducing slower AF or flutter with 1:1 ventricular conduction is presumably comparable to that of these agents and remains exceedingly rare, occurring only in isolated cases, and is typically a transient occurrence during a stage of slowed AF or flutter.[60,61]

Currently, synchronized direct-current cardioversion remains the first choice for acute treatment, particularly in hemodynamically unstable cases. However, amiodarone may be considered as a safe alternative, despite failing to cardiovert in some cases. This is particularly significant in countries highlighted by our survey, where amiodarone is the only intravenous antiarrhythmic available for emergency rhythm control. Importantly, during pharmacological cardioversion of WPW patients with any agent possessing potent blockade of the sodium channel, including amiodarone, there should be close monitoring of the patient with a bedside defibrillator until one hour after successful restoration of sinus rhythm, due to the small risk of degeneration into VF (Figure 4).

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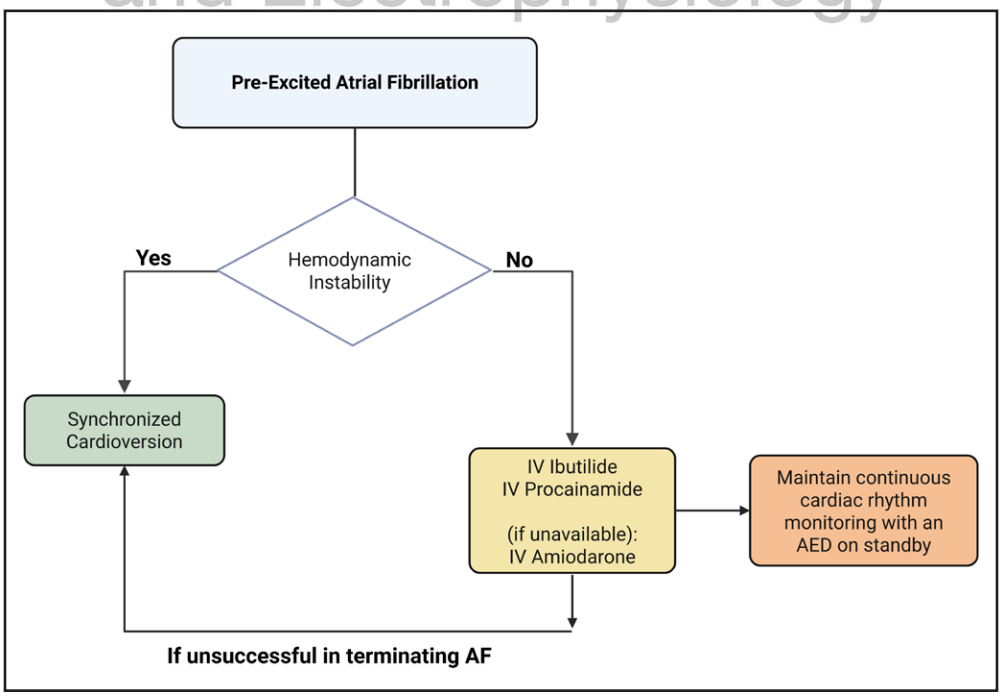


**Figure 4. Algorithm for acute therapy of preexcited atrial fibrillation (AF).**
In hemodynamically stable patients, intravenous (IV) ibutilide and IV procainamide are recommended as the preferred agents for pharmacological cardioversion. If unavailable, IV amiodarone may safely be used with close monitoring from administration to one hour following conversion to sinus rhythm. AED indicates automated external defibrillator.

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## Limitations

There are several limitations that should be acknowledged in interpreting our findings. First, despite a systematic approach, the available population-level evidence is all single-arm observational and interventional. The absence of control arms and the small sample sizes limit the ability to infer causality and preclude reliable meta-analysis. As a reference, a minimum of 600 cases would be needed to constrain the upper-bound on risk below 0.5% by concordant Beta–binomial bounds.

Second, case reports are inherently subject to reporting bias since adverse events are more likely to be given greater clinical attention in anecdotal literature as opposed to uneventful outcomes, which inflates perceived risk. Third, heterogeneity in amiodarone dosing and in baseline patient characteristics (particularly the location of the AP, preexisting cardiovascular conditions, and AP-AERP) was variably reported across studies and limits generalizability; because events were rare, formal heterogeneity estimation was not informative, and pooled estimates should be interpreted as conservative bounds.

Fourth, the survey of antiarrhythmic availability represents a convenience sample with a modest response rate (10 of 27 centers) and therefore may be subject to selection and nonresponse bias, respectively; accordingly, these findings should not be generalized to regional (let alone worldwide) availability.

## Conclusions and Future Directions

In this systematic review, we found an absence of definitive evidence regarding the risk profile of amiodarone in preexcited AF. Population-level data suggest a low risk of VF for amiodarone. Given the prevalent use of amiodarone among many hospitals in developing countries, as per the survey results, there is a need for a randomized control trial and a multi-center registry that tracks outcomes of amiodarone administration for preexcited AF cases among these centers.

## ARTICLE INFORMATION

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